

EW – 1 Lab 1

Harshit Goyal

2023102054

Saarthak Sabharwal

2023102055

INTRODUCTION TO DSO AND BREADBOARD:

An oscilloscope is a device used to analyse an (electric) signal graphically. Early oscilloscopes (Cathode Ray Oscilloscopes, CROs) had a Cathode Ray Tube (CRT) within which worked with a beam of electrons. Instead of cathode rays, modern oscilloscopes (called Digital Storage Oscilloscopes, DSOs) use digitized methods to process and reconstruct a waveform on a screen and store it as well.

An RC circuit consists of only resistors and capacitors. The simplest RC circuit is a series-connected resistor and capacitor. We will provide sinusoidal (AC) input to this circuit and observe the output voltage across the capacitor, via the DSO.

Cutoff frequency, $f_c = \frac{1}{2\pi RC}$.

AIM: To analyse voltage across capacitor on applying AC voltage to RC circuit.

Components Required:

- Resistor
- Capacitor
- Breadboard
- Wires
- DSO

Procedure:

- Connect the resistor and capacitor in series on the breadboard. Provide input to the circuit and output across the capacitor via wires.
- Apply sine wave using the function generator.
- Note down the voltage across capacitor at $t=RC$ (time constant). Tabulate these values against varying frequencies.

Results:

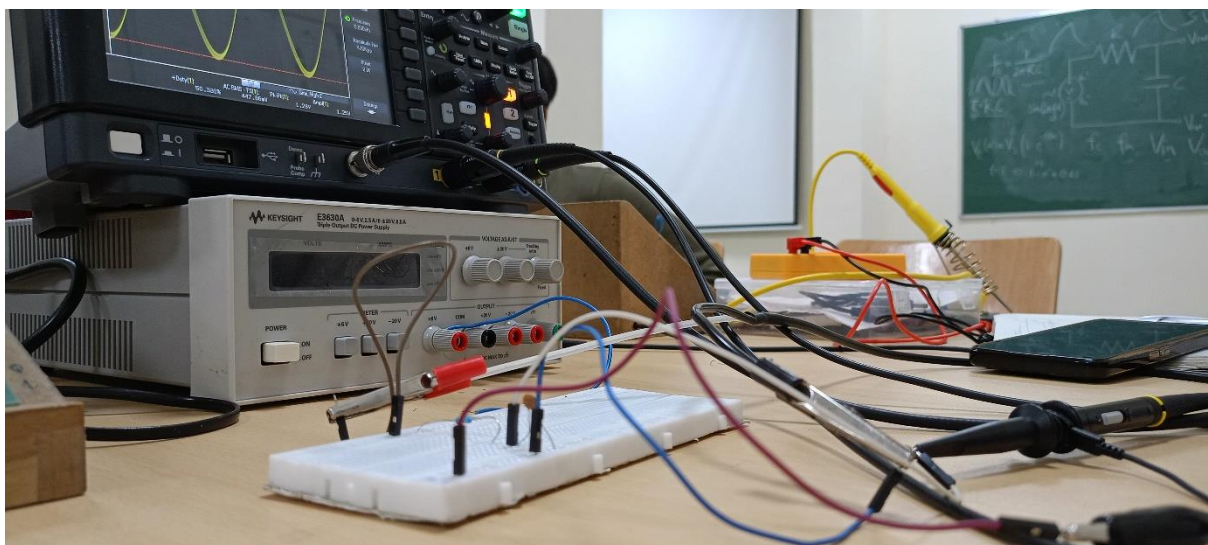
$$R = 1000 \, \Omega$$

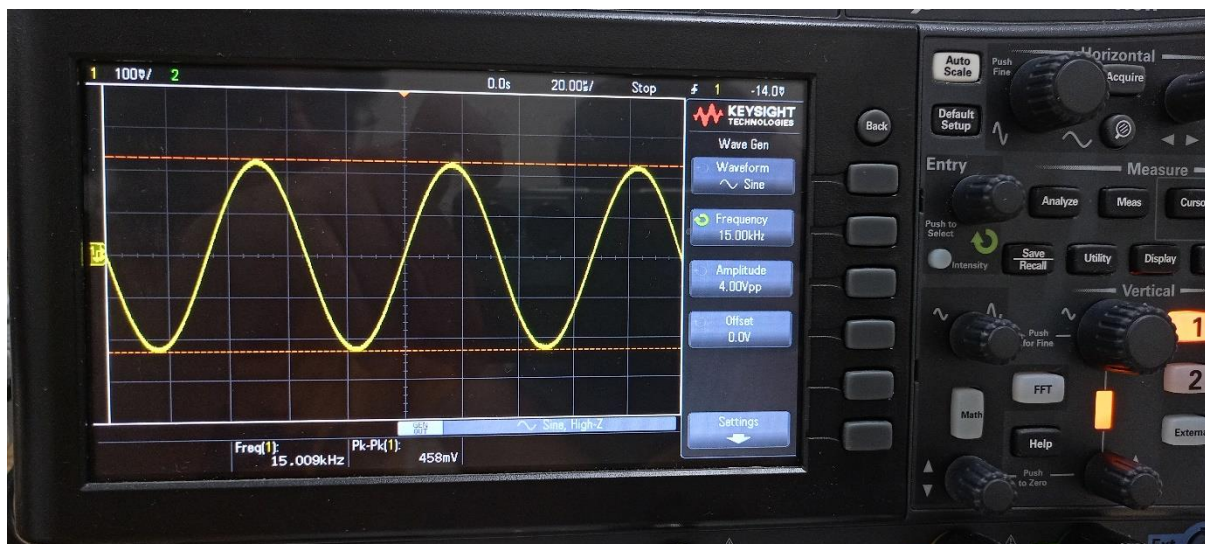
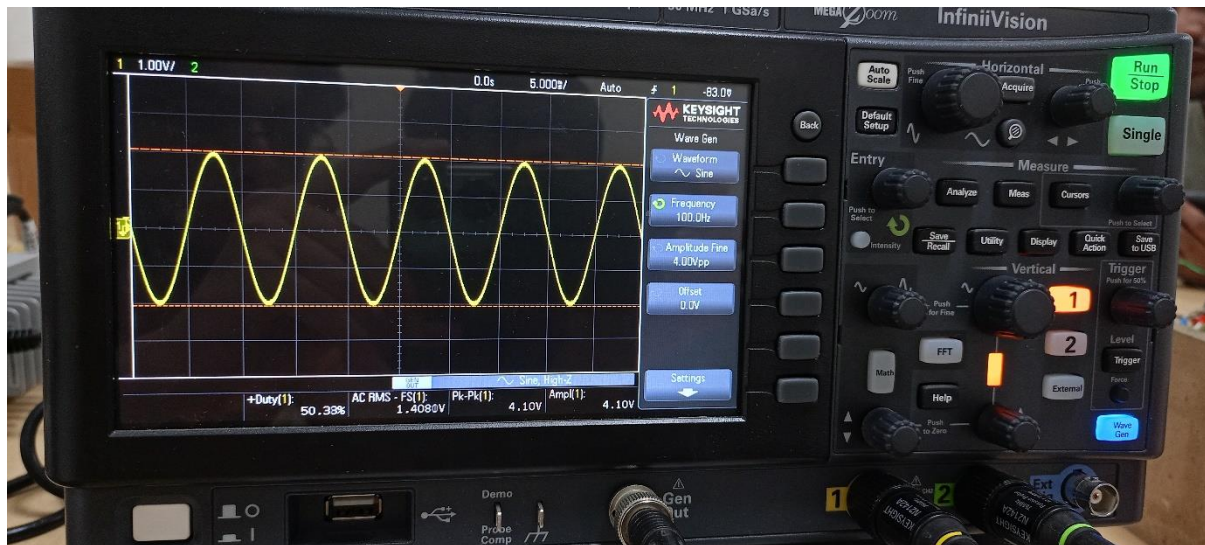
$$C = 0.1 \, \mu\text{F}$$

$$f_c = 1591 \, \text{Hz}$$

$$V_{in} = 4\text{V}$$

S.No.	Frequency (Hz)	V_{out} (V)
1.	100	4
2.	300	3.94
3.	500	3.74
4.	1200	2.89
5.	1600	2.37
6.	2000	2.05
7.	3000	1.75
8.	5000	1.17
9.	10000	0.65
10.	15000	0.46





Observations:

The voltage across the capacitor (at $t = RC$) drops gradually at first, due to accumulation of charge. Around the cutoff frequency, this voltage drops steeply to a certain value, after which the slope graduals out again. Errors in the recordings are likely to arise due to poor connections and/or grips.

Conclusion: The DSO was successfully used to read and generate all sorts of outputs.